Coverage Report by instance with details

```
================================================================================
=
=== Instance: /spi_master_top/u_dut/u_regfile/u_apb_sva
=== Design Unit: work.apb_regfile_sva
================================================================================
=

Assertion Coverage:
    Assertions                      13        13         0   100.00%
--------------------------------------------------------------------
Name                 File(Line)                   Failure      Pass
                                                  Count        Count
--------------------------------------------------------------------
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__irq_reset_value
                     ../tb/assertions/spi_sva.sv(210)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__int_stat_reset_value
                     ../tb/assertions/spi_sva.sv(199)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__ctrl_en_reset_value
                     ../tb/assertions/spi_sva.sv(188)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_drop_implies_full
                     ../tb/assertions/spi_sva.sv(177)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_push_not_when_full
                     ../tb/assertions/spi_sva.sv(166)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__rx_ovf_sets_int_stat
                     ../tb/assertions/spi_sva.sv(155)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_ovf_sets_int_stat
                     ../tb/assertions/spi_sva.sv(144)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__irq_equation_correct
                     ../tb/assertions/spi_sva.sv(132)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_pslverr_always_0
                     ../tb/assertions/spi_sva.sv(121)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_pready_always_1
                     ../tb/assertions/spi_sva.sv(110)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/
assert__apb_ctrl_stable_setup_to_access
                     ../tb/assertions/spi_sva.sv(99)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_penable_requires_psel
                     ../tb/assertions/spi_sva.sv(78)
                                                        0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_psel_min_2_cycles
                     ../tb/assertions/spi_sva.sv(67)
                                                        0          1

Directive Coverage:
    Directives                      13        13         0   100.00%

DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------
------------
Name                                     Design Design   Lang File(Line)
Hits Status
```

```
                                         Unit   UnitType
--------------------------------------------------------------------------------
------------
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__irq_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(212)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__int_stat_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(201)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__ctrl_en_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(190)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_drop_implies_full
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(179)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_push_not_when_full
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(168)

21262 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__rx_ovf_sets_int_stat
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(157)

3 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_ovf_sets_int_stat
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(146)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__irq_equation_correct
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(135)

21262 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_pslverr_always_0
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(123)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_pready_always_1
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(112)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_ctrl_stable_setup_to_access

                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(101)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_penable_requires_psel
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(80)
```

```
10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_psel_min_2_cycles
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(69)

6 Covered

================================================================================
=
=== Instance: /spi_master_top/u_dut/u_regfile
=== Design Unit: work.apb_regfile
================================================================================
=
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        55        55         0   100.00%

================================Branch Details================================

Branch Coverage for instance /spi_master_top/u_dut/u_regfile

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ../rtl/apb_regfile.sv
------------------------------------IF
Branch------------------------------------
    170                                    21399     Count coming in to IF
    170              1                     10361             if (apb_read) begin
                                           11038     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------CASE
Branch------------------------------------
    171                                    10361     Count coming in to CASE
    172              1                         5                     OFF_CTRL
: PRDATA = ctrl_word;
    173              1                     10172                     OFF_STATUS
: PRDATA = status_word;
    174              1                         1                     OFF_TX_DATA
: PRDATA = 32'h0;                // WO
    175              1                       114                     OFF_RX_DATA
: begin
    179              1                         9                     OFF_CLK_DIV
: PRDATA = clk_div_word;
    180              1                        12                     OFF_SS_CTRL
: PRDATA = ss_ctrl_word;
    181              1                         5                     OFF_INT_EN
: PRDATA = int_en_word;
    182              1                        33
OFF_INT_STAT: PRDATA = int_stat_word;
    183              1                         5                     OFF_DELAY
: PRDATA = delay_word;
    184              1                         5                     default
: PRDATA = 32'h0;                // R23
Branch totals: 10 hits of 10 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    176                                      114     Count coming in to IF
    176              1                        40                         PRDATA
= rx_empty_w ? 32'h0 : rx_mem[rx_rp[FIFO_AW-1:0]];
    176              2                        74                         PRDATA
= rx_empty_w ? 32'h0 : rx_mem[rx_rp[FIFO_AW-1:0]];
```

```
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    198                                     1137     Count coming in to IF
    198             1                         99             if (apb_write &&
PADDR == OFF_TX_DATA && ctrl_en) begin
                                            1038     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------CASE
Branch------------------------------------
    200                                       99     Count coming in to CASE
    201             1                         83                     2'b00:
tx_push_data = {24'b0, PWDATA[7:0]};
    202             1                          8                     2'b01:
tx_push_data = {16'b0, PWDATA[15:0]};
    203             1                          8                     default:
tx_push_data = PWDATA;
Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    215                                     1307     Count coming in to IF
    215             1                         12             if (!PRESETn) begin
    228             1                       1295             end else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    229                                     1295     Count coming in to IF
    229             1                        428                 if (apb_write)
begin
                                             867     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------CASE
Branch------------------------------------
    230                                      428     Count coming in to CASE
    231             1                         98
OFF_CTRL: begin
    239             1                         48
OFF_CLK_DIV: clk_div <= PWDATA[15:0];
    240             1                        103
OFF_SS_CTRL: begin
    244             1                         14
OFF_INT_EN : int_en    <= PWDATA[IRQ_COUNT-1:0];
    245             1                         11
OFF_DELAY  : delay_cfg <= PWDATA[7:0];
    246             1                        154
default: ;
Branch totals: 6 hits of 6 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    258                                     1295     Count coming in to IF
    258             1                         53                     if
(apb_write && PADDR == OFF_INT_STAT) begin
                                            1242     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    262                                     1295     Count coming in to IF
```

```
    262            1                            6                     if
(tx_push_dropped)
                                             1289     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    264                                      1295     Count coming in to IF
    264            1                            3                     if
(rx_push_valid && rx_full_w)
                                             1292     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    266                                      1295     Count coming in to IF
    266            1                            2                     if
(rx_push_valid && !rx_full_w && (rx_count == FIFO_DEPTH-1))
                                             1293     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    268                                      1295     Count coming in to IF
    268            1                           47                     if (tx_pop
&& (tx_count == 1))
                                             1248     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    270                                      1295     Count coming in to IF
    270            1                           65                     if
(transfer_done_pulse)
                                             1230     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    286                                       461     Count coming in to IF
    286            1                           12             if (!PRESETn) begin
    290            1                           87             end else if (!
ctrl_en) begin
    293            1                          362             end else begin
Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    294                                       362     Count coming in to IF
    294            1                           93                 if
(tx_push_accepted) begin
                                              269     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    298                                       362     Count coming in to IF
    298            1                           68                 if (tx_pop)
begin
                                              294     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
```

```
    308                                     376     Count coming in to IF
    308             1                        12             if (!PRESETn) begin
    312             1                        82             end else if (!
ctrl_en) begin
    315             1                       282             end else begin
Branch totals: 3 hits of 3 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    316                                     282     Count coming in to IF
    316             1                        62                 if
(rx_push_valid && !rx_full_w) begin
                                            220     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    320                                     282     Count coming in to IF
    320             1                        55                 if
(rx_pop_this_cycle) begin
                                            227     All False Count
Branch totals: 2 hits of 2 branches = 100.00%


Condition Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Conditions                      14        14         0   100.00%

================================Condition
Details================================

Condition Coverage for instance /spi_master_top/u_dut/u_regfile --

  File ../rtl/apb_regfile.sv
----------------Focused Condition View-------------------
Line       198 Item    1  ((apb_write && (PADDR == 8)) && ctrl_en)
Condition totals: 3 of 3 input terms covered = 100.00%

    Input Term   Covered  Reason for no coverage   Hint
   -----------  --------  -----------------------  --------------
     apb_write         Y
  (PADDR == 8)         Y
       ctrl_en         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  apb_write_0           -
  Row   2:          1  apb_write_1           (ctrl_en && (PADDR == 8))
  Row   3:          1  (PADDR == 8)_0        apb_write
  Row   4:          1  (PADDR == 8)_1        (ctrl_en && apb_write)
  Row   5:          1  ctrl_en_0             (apb_write && (PADDR == 8))
  Row   6:          1  ctrl_en_1             (apb_write && (PADDR == 8))

----------------Focused Condition View-------------------
Line       258 Item    1  (apb_write && (PADDR == 28))
Condition totals: 2 of 2 input terms covered = 100.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  -----------------------  --------------
      apb_write         Y
  (PADDR == 28)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
```

```
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  apb_write_0           -
  Row   2:          1  apb_write_1           (PADDR == 28)
  Row   3:          1  (PADDR == 28)_0       apb_write
  Row   4:          1  (PADDR == 28)_1       apb_write

----------------Focused Condition View-------------------
Line       264 Item    1  (rx_push_valid && rx_full_w)
Condition totals: 2 of 2 input terms covered = 100.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
  rx_push_valid         Y
      rx_full_w         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  rx_push_valid_0       -
  Row   2:          1  rx_push_valid_1       rx_full_w
  Row   3:          1  rx_full_w_0           rx_push_valid
  Row   4:          1  rx_full_w_1           rx_push_valid

----------------Focused Condition View-------------------
Line       266 Item    1  ((rx_push_valid && ~rx_full_w) && (rx_count == (8 -
1)))
Condition totals: 3 of 3 input terms covered = 100.00%

             Input Term   Covered  Reason for no coverage   Hint
            -----------  --------  ----------------------  --------------
          rx_push_valid         Y
              rx_full_w         Y
  (rx_count == (8 - 1))         Y

     Rows:       Hits  FEC Target               Non-masking condition(s)
 ---------  ---------  --------------------     -------------------------
  Row   1:          1  rx_push_valid_0          -
  Row   2:          1  rx_push_valid_1          ((rx_count == (8 - 1)) &&
~rx_full_w)
  Row   3:          1  rx_full_w_0              ((rx_count == (8 - 1)) &&
rx_push_valid)
  Row   4:          1  rx_full_w_1              rx_push_valid
  Row   5:          1  (rx_count == (8 - 1))_0  (rx_push_valid && ~rx_full_w)
  Row   6:          1  (rx_count == (8 - 1))_1  (rx_push_valid && ~rx_full_w)

----------------Focused Condition View-------------------
Line       268 Item    1  (tx_pop && (tx_count == 1))
Condition totals: 2 of 2 input terms covered = 100.00%

       Input Term   Covered  Reason for no coverage   Hint
      -----------  --------  ----------------------  --------------
           tx_pop         Y
  (tx_count == 1)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  tx_pop_0              -
  Row   2:          1  tx_pop_1              (tx_count == 1)
  Row   3:          1  (tx_count == 1)_0     tx_pop
  Row   4:          1  (tx_count == 1)_1     tx_pop

----------------Focused Condition View-------------------
Line       316 Item    1  (rx_push_valid && ~rx_full_w)
Condition totals: 2 of 2 input terms covered = 100.00%
```

```
     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
  rx_push_valid         Y
      rx_full_w         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  rx_push_valid_0       -
  Row   2:          1  rx_push_valid_1       ~rx_full_w
  Row   3:          1  rx_full_w_0           rx_push_valid
  Row   4:          1  rx_full_w_1           rx_push_valid


Expression Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Expressions                     14        13         1    92.85%

================================Expression
Details================================

Expression Coverage for instance /spi_master_top/u_dut/u_regfile --

  File ../rtl/apb_regfile.sv
----------------Focused Expression View-----------------
Line       109 Item    1  (PSEL & PENABLE)
Expression totals: 1 of 2 input terms covered = 50.00%

  Input Term   Covered  Reason for no coverage   Hint
 -----------  --------  ----------------------  --------------
        PSEL         N  '_0' not hit             Hit '_0'
     PENABLE         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:    ***0***  PSEL_0                PENABLE
  Row   2:          1  PSEL_1                PENABLE
  Row   3:          1  PENABLE_0             PSEL
  Row   4:          1  PENABLE_1             PSEL

----------------Focused Expression View-----------------
Line       110 Item    1  (apb_access & PWRITE)
Expression totals: 2 of 2 input terms covered = 100.00%

  Input Term   Covered  Reason for no coverage   Hint
 -----------  --------  ----------------------  --------------
  apb_access         Y
      PWRITE         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  apb_access_0          PWRITE
  Row   2:          1  apb_access_1          PWRITE
  Row   3:          1  PWRITE_0              apb_access
  Row   4:          1  PWRITE_1              apb_access

----------------Focused Expression View-----------------
Line       111 Item    1  (apb_access & ~PWRITE)
Expression totals: 2 of 2 input terms covered = 100.00%

  Input Term   Covered  Reason for no coverage   Hint
 -----------  --------  ----------------------  --------------
  apb_access         Y
      PWRITE         Y
```

```
     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  apb_access_0          ~PWRITE
  Row   2:          1  apb_access_1          ~PWRITE
  Row   3:          1  PWRITE_0              apb_access
  Row   4:          1  PWRITE_1              apb_access

----------------Focused Expression View-----------------
Line       124 Item    1  (tx_count == 8)
Expression totals: 1 of 1 input term covered = 100.00%

       Input Term   Covered  Reason for no coverage   Hint
      -----------  --------  ----------------------  --------------
  (tx_count == 8)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (tx_count == 8)_0     -
  Row   2:          1  (tx_count == 8)_1     -

----------------Focused Expression View-----------------
Line       125 Item    1  (tx_count == 0)
Expression totals: 1 of 1 input term covered = 100.00%

       Input Term   Covered  Reason for no coverage   Hint
      -----------  --------  ----------------------  --------------
  (tx_count == 0)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (tx_count == 0)_0     -
  Row   2:          1  (tx_count == 0)_1     -

----------------Focused Expression View-----------------
Line       130 Item    1  (rx_count == 8)
Expression totals: 1 of 1 input term covered = 100.00%

       Input Term   Covered  Reason for no coverage   Hint
      -----------  --------  ----------------------  --------------
  (rx_count == 8)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (rx_count == 8)_0     -
  Row   2:          1  (rx_count == 8)_1     -

----------------Focused Expression View-----------------
Line       131 Item    1  (rx_count == 0)
Expression totals: 1 of 1 input term covered = 100.00%

       Input Term   Covered  Reason for no coverage   Hint
      -----------  --------  ----------------------  --------------
  (rx_count == 0)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (rx_count == 0)_0     -
  Row   2:          1  (rx_count == 0)_1     -

----------------Focused Expression View-----------------
Line       208 Item    1  (tx_push_valid & ~tx_full_w)
Expression totals: 2 of 2 input terms covered = 100.00%
```

```
     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
  tx_push_valid         Y
      tx_full_w         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  tx_push_valid_0       ~tx_full_w
  Row   2:          1  tx_push_valid_1       ~tx_full_w
  Row   3:          1  tx_full_w_0           tx_push_valid
  Row   4:          1  tx_full_w_1           tx_push_valid

----------------Focused Expression View-----------------
Line       209 Item    1  (tx_push_valid & tx_full_w)
Expression totals: 2 of 2 input terms covered = 100.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
  tx_push_valid         Y
      tx_full_w         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  tx_push_valid_0       tx_full_w
  Row   2:          1  tx_push_valid_1       tx_full_w
  Row   3:          1  tx_full_w_0           tx_push_valid
  Row   4:          1  tx_full_w_1           tx_push_valid


Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      96        96         0   100.00%

================================Statement
Details================================

Statement Coverage for instance /spi_master_top/u_dut/u_regfile --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ../rtl/apb_regfile.sv
    25                                               module apb_regfile (
    26                                                   input  wire
PCLK,
    27                                                   input  wire
PRESETn,
    28
    29                                                   // APB slave
    30                                                   input  wire
PSEL,
    31                                                   input  wire
PENABLE,
    32                                                   input  wire
PWRITE,
    33                                                   input  wire [7:0]
PADDR,
    34                                                   input  wire [31:0]
PWDATA,
    35                                                   output reg  [31:0]
PRDATA,
    36                                                   output wire
PREADY,
    37                                                   output wire
```

```
PSLVERR,

// Configuration to core
output wire cfg_en,
output wire cfg_mstr,
output wire [1:0] cfg_mode,
output wire cfg_lsb_first,
output wire cfg_loopback,
output wire [1:0] cfg_width,
output wire [15:0] cfg_clk_div,
output wire [7:0] cfg_delay,

// SS lane drive (combinational, outputs via SS_n)
output wire [3:0] SS_n,

// TX FIFO interface to core
output wire [31:0] tx_word,        // head of TX FIFO
output wire tx_empty,
input  wire tx_pop,         // core pops one word

// RX FIFO interface from core
input  wire rx_push_valid,
input  wire [31:0] rx_push_data,

// Core status -> regfile
input  wire busy_in,
input  wire transfer_done_pulse,

// Aggregate interrupt
output wire         IRQ
);

// ----------------------------------------------------------------------
// Register offsets
// ----------------------------------------------------------------------
localparam [7:0] OFF_CTRL     = 8'h00;
localparam [7:0] OFF_STATUS   = 8'h04;
localparam [7:0] OFF_TX_DATA  = 8'h08;
```

```
                                     localparam [7:0] OFF_RX_DATA  = 8'h0C;
                                     localparam [7:0] OFF_CLK_DIV  = 8'h10;
                                     localparam [7:0] OFF_SS_CTRL  = 8'h14;
                                     localparam [7:0] OFF_INT_EN   = 8'h18;
                                     localparam [7:0] OFF_INT_STAT = 8'h1C;
                                     localparam [7:0] OFF_DELAY    = 8'h20;

                                     localparam integer IRQ_TX_EMPTY      = 0;
                                     localparam integer IRQ_RX_FULL       = 1;
                                     localparam integer IRQ_TX_OVF        = 2;
                                     localparam integer IRQ_RX_OVF        = 3;
                                     localparam integer IRQ_TRANSFER_DONE = 4;
                                     localparam integer IRQ_COUNT         = 5;

                                     // ----------------------------------------------------------------------------
                                     // Configuration registers
                                     // ----------------------------------------------------------------------------
                                     reg        ctrl_en;
                                     reg        ctrl_mstr;
                                     reg [1:0]  ctrl_mode; // {CPOL, CPHA}
                                     reg        ctrl_lsb_first;
                                     reg        ctrl_loopback;
                                     reg [1:0]  ctrl_width;

                                     reg [15:0] clk_div;
                                     reg [3:0]  ss_en;
                                     reg [3:0]  ss_val;
                                     reg [IRQ_COUNT-1:0] int_en;
                                     reg [IRQ_COUNT-1:0] int_stat;
                                     reg [7:0]  delay_cfg;

                                     // ----------------------------------------------------------------------------
                                     // APB handshake
                                     // ----------------------------------------------------------------------------
1          21269                     wire apb_access = PSEL & PENABLE;
1          21268                     wire apb_write  = apb_access &  PWRITE;
1          21268                     wire apb_read   = apb_access & ~PWRITE;
                                     assign PREADY   = 1'b1;
                                     assign PSLVERR  = 1'b0;
```

```
114
115                                      // -------------------------------------------------------------------------
116                                      // FIFO storage - 8 deep, 32 wide
117                                      // -------------------------------------------------------------------------
118                                      localparam integer FIFO_DEPTH = 8;
119                                      localparam integer FIFO_AW    = 3;
120
121                                      reg [31:0] tx_mem [0:FIFO_DEPTH-1];
122                                      reg [FIFO_AW:0] tx_wp, tx_rp;
123         1                198         wire [FIFO_AW:0] tx_count = tx_wp - tx_rp;
124         1                155         wire tx_full_w  = (tx_count == FIFO_DEPTH);
125         1                155         wire tx_empty_w = (tx_count == 0);
126
127                                      reg [31:0] rx_mem [0:FIFO_DEPTH-1];
128                                      reg [FIFO_AW:0] rx_wp, rx_rp;
129         1                155         wire [FIFO_AW:0] rx_count = rx_wp - rx_rp;
130         1                120         wire rx_full_w  = (rx_count == FIFO_DEPTH);
131         1                120         wire rx_empty_w = (rx_count == 0);
132
133                                      assign tx_empty = tx_empty_w;
134         1                182         assign tx_word  = tx_mem[tx_rp[FIFO_AW-1:0]];
135
136                                      // -------------------------------------------------------------------------
137                                      // Combinational APB read data
138                                      // -------------------------------------------------------------------------
139         1                406         wire [31:0] status_word = {
140                                          25'b0,
141 int_stat[IRQ_RX_OVF],     // [6] RX_OVF
142 int_stat[IRQ_TX_OVF],     // [5] TX_OVF
143                                          rx_empty_w, // [4] RX_EMPTY (reset = 1)
144                                          rx_full_w, // [3] RX_FULL
145                                          tx_empty_w, // [2] TX_EMPTY (reset = 1)
146                                          tx_full_w, // [1] TX_FULL
147                                          busy_in // [0] BUSY
148                                      };
149
```

```
    150            1                          92         wire [31:0] ctrl_word = {
    151                                                          24'b0,
    152                                                          ctrl_width,
    153                                                          ctrl_loopback,
    154                                                          ctrl_lsb_first,
    155                                                          ctrl_mode,
    156                                                          ctrl_mstr,
    157                                                          ctrl_en
    158                                                      };
    159
    160            1                         100         wire [31:0] ss_ctrl_word  = {24'b0, ss_val,    ss_en};
    161            1                          12         wire [31:0] int_en_word = {{(32-IRQ_COUNT){1'b0}}, int_en};
    162            1                         141         wire [31:0] int_stat_word = {{(32-IRQ_COUNT){1'b0}}, int_stat};
    163            1                          28         wire [31:0] clk_div_word  = {16'b0, clk_div};
    164            1                          11         wire [31:0] delay_word = {24'b0, delay_cfg};
    165
    166                                                  reg rx_pop_this_cycle;
    167            1                       21399         always @(*) begin
    168            1                       21399             rx_pop_this_cycle = 1'b0;
    169            1                       21399             PRDATA = 32'h0;
    170                                                      if (apb_read) begin
    171                                                          case (PADDR)
    172            1                           5                     OFF_CTRL : PRDATA = ctrl_word;
    173            1                       10172                     OFF_STATUS : PRDATA = status_word;
    174            1                           1                     OFF_TX_DATA : PRDATA = 32'h0;                // WO
    175                                                              OFF_RX_DATA : begin
    176            1                         114                         PRDATA = rx_empty_w ? 32'h0 : rx_mem[rx_rp[FIFO_AW-1:0]];
    177            1                         114 rx_pop_this_cycle = ~rx_empty_w;         // R15: empty read no OVF
    178                                                              end
    179            1                           9                     OFF_CLK_DIV : PRDATA = clk_div_word;
    180            1                          12                     OFF_SS_CTRL : PRDATA = ss_ctrl_word;
    181            1                           5                     OFF_INT_EN : PRDATA = int_en_word;
    182            1                          33 OFF_INT_STAT: PRDATA = int_stat_word;
    183            1                           5                     OFF_DELAY : PRDATA = delay_word;
    184            1                           5                     default : PRDATA = 32'h0;                // R23
    185                                                          endcase
    186                                                      end
    187                                                  end
    188
    189                                                  // --------------------------------------------------------------------------
    190                                                  // TX push side
    191                                                  // --------------------------------------------------------------------------
    192                                                  reg tx_push_valid;
```

```
    193                                              reg [31:0] tx_push_data;
    194
    195        1                      1137           always @(*) begin
    196        1                      1137               tx_push_valid = 1'b0;
    197        1                      1137               tx_push_data  = 32'h0;
    198                                                  if (apb_write && PADDR == OFF_TX_DATA && ctrl_en) begin
    199        1                        99                   tx_push_valid = 1'b1;
    200                                                      case (ctrl_width)
    201        1                        83                       2'b00: tx_push_data = {24'b0, PWDATA[7:0]};
    202        1                         8                       2'b01: tx_push_data = {16'b0, PWDATA[15:0]};
    203        1                         8                       default: tx_push_data = PWDATA;
    204                                                      endcase
    205                                                  end
    206                                              end
    207
    208        1                       209           wire tx_push_accepted = tx_push_valid & ~tx_full_w;
    209        1                       209           wire tx_push_dropped  = tx_push_valid &  tx_full_w;   // sets TX_OVF
    210
    211                                              // -------------------------------------------------------------------------
    212                                              // Configuration register bank
    213                                              // -------------------------------------------------------------------------
    214        1                      1307           always @(posedge PCLK or negedge PRESETn) begin
    215                                                  if (!PRESETn) begin
    216        1                        12                   ctrl_en <= 1'b0;
    217        1                        12                   ctrl_mstr <= 1'b0;
    218        1                        12                   ctrl_mode <= 2'b00;
    219        1                        12                   ctrl_lsb_first <= 1'b0;
    220        1                        12                   ctrl_loopback <= 1'b0;
    221        1                        12                   ctrl_width <= 2'b00;
    222        1                        12                   clk_div <= 16'h0;
    223        1                        12                   ss_en <= 4'h0;
    224        1                        12                   ss_val <= 4'h0;
    225        1                        12                   int_en <= '0;
    226        1                        12                   int_stat <= '0;
    227        1                        12                   delay_cfg <= 8'h0;
    228                                                  end else begin
    229                                                      if (apb_write)
```

```
begin
    230                                                              case
(PADDR)
    231
OFF_CTRL: begin
    232            1                          98
ctrl_width     <= PWDATA[7:6];
    233            1                          98
ctrl_loopback  <= PWDATA[5];
    234            1                          98
ctrl_lsb_first <= PWDATA[4];
    235            1                          98
ctrl_mode      <= PWDATA[3:2];
    236            1                          98
ctrl_mstr      <= PWDATA[1];
    237            1                          98
ctrl_en        <= PWDATA[0];
    238                                                                  end
    239            1                          48
OFF_CLK_DIV: clk_div <= PWDATA[15:0];
    240
OFF_SS_CTRL: begin
    241            1                         103
ss_val <= PWDATA[7:4];
    242            1                         103
ss_en  <= PWDATA[3:0];
    243                                                                  end
    244            1                          14
OFF_INT_EN : int_en    <= PWDATA[IRQ_COUNT-1:0];
    245            1                          11
OFF_DELAY  : delay_cfg <= PWDATA[7:0];
    246
default: ;
    247                                                              endcase
    248                                                          end
    249
    250                                                          // -----
INT_STAT update (R18 W1C priority) -----------------------
    251                                                          //   1)
default: hold value
    252                                                          //   2) apply
W1C mask
    253                                                          //   3) OR in
new events (so set+clear stays set)
    254                                                          begin :
int_stat_update
    255                                                              reg
[IRQ_COUNT-1:0] next_stat;
    256            1                        1295                     next_stat =
int_stat;
    257
    258                                                              if
(apb_write && PADDR == OFF_INT_STAT) begin
    259            1                          53
next_stat = next_stat & ~PWDATA[IRQ_COUNT-1:0];
    260                                                              end
    261
    262                                                              if
(tx_push_dropped)
    263            1                           6
next_stat[IRQ_TX_OVF] = 1'b1;
    264                                                              if
(rx_push_valid && rx_full_w)
    265            1                           3
```

```
next_stat[IRQ_RX_OVF] = 1'b1;
    266                                                                if
(rx_push_valid && !rx_full_w && (rx_count == FIFO_DEPTH-1))
    267            1                         2
next_stat[IRQ_RX_FULL] = 1'b1;
    268                                                                if (tx_pop
&& (tx_count == 1))
    269            1                        47
next_stat[IRQ_TX_EMPTY] = 1'b1;
    270                                                                if
(transfer_done_pulse)
    271            1                        65
next_stat[IRQ_TRANSFER_DONE] = 1'b1;
    272
    273            1                      1295                         int_stat <=
next_stat;
    274                                                            end
    275                                                        end
    276                                                    end
    277
    278            1                       151         assign IRQ  = |
(int_stat & int_en);
    279            1                       100         assign SS_n = ~ss_en |
ss_val;    // R20
    280
    281                                                //
-----------------------------------------------------------------------
    282                                                // TX FIFO storage
    283                                                //
-----------------------------------------------------------------------
    284                                                integer i;
    285            1                       461         always @(posedge PCLK
or negedge PRESETn) begin
    286                                                    if (!PRESETn) begin
    287            1                        12                 tx_wp <= '0;
    288            1                        12                 tx_rp <= '0;
    289            1                        12                 for (i = 0; i <
FIFO_DEPTH; i = i + 1) tx_mem[i] <= 32'h0;
    289            2                        96
    289            3                        96
    290                                                    end else if (!
ctrl_en) begin
    291            1                        87                 tx_wp <= '0;
    292            1                        87                 tx_rp <= '0;
    293                                                    end else begin
    294                                                        if
(tx_push_accepted) begin
    295            1                        93
tx_mem[tx_wp[FIFO_AW-1:0]] <= tx_push_data;
    296            1                        93                     tx_wp <=
tx_wp + 1'b1;
    297                                                        end
    298                                                        if (tx_pop)
begin
    299            1                        68                     tx_rp <=
tx_rp + 1'b1;
    300                                                        end
    301                                                    end
    302                                                end
    303
    304                                                //
-----------------------------------------------------------------------
    305                                                // RX FIFO storage
    306                                                //
```

```
---------------------------------------------------------------------------
    307             1                        376         always @(posedge PCLK
or negedge PRESETn) begin
    308                                                      if (!PRESETn) begin
    309             1                         12                 rx_wp <= '0;
    310             1                         12                 rx_rp <= '0;
    311             1                         12                 for (i = 0; i <
FIFO_DEPTH; i = i + 1) rx_mem[i] <= 32'h0;
    311             2                         96
    311             3                         96
    312                                                      end else if (!
ctrl_en) begin
    313             1                         82                 rx_wp <= '0;
    314             1                         82                 rx_rp <= '0;
    315                                                      end else begin
    316                                                          if
(rx_push_valid && !rx_full_w) begin
    317             1                         62
rx_mem[rx_wp[FIFO_AW-1:0]] <= rx_push_data;
    318             1                         62                     rx_wp <=
rx_wp + 1'b1;
    319                                                          end
    320                                                          if
(rx_pop_this_cycle) begin
    321             1                         55                     rx_rp <=
rx_rp + 1'b1;

Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                       1116       714       402    63.97%

================================Toggle Details================================

Toggle Coverage for instance /spi_master_top/u_dut/u_regfile --

                                              Node      1H->0L      0L->1H
"Coverage"

---------------------------------------------------------------
                                               IRQ           1           1
100.00
                                        PADDR[0-7]           1           1
100.00
                                              PCLK           1           1
100.00
                                           PENABLE           1           1
100.00
                                      PRDATA[31-0]           1           1
100.00
                                            PREADY           0           0
0.00
                                           PRESETn           1           1
100.00
                                              PSEL           1           1
100.00
                                           PSLVERR           0           0
0.00
                                      PWDATA[0-31]           1           1
100.00
                                            PWRITE           1           1
100.00
                                         SS_n[0-3]           1           1
100.00
```

apb_access	1	1	100.00
apb_read	1	1	100.00
apb_write	1	1	100.00
busy_in	1	1	100.00
cfg_clk_div[0-15]	1	1	100.00
cfg_delay[0-7]	1	1	100.00
cfg_en	1	1	100.00
cfg_loopback	1	1	100.00
cfg_lsb_first	1	1	100.00
cfg_mode[0-1]	1	1	100.00
cfg_mstr	1	1	100.00
cfg_width[0-1]	1	1	100.00
clk_div[15-0]	1	1	100.00
clk_div_word[0-15]	1	1	100.00
clk_div_word[16-31]	0	0	0.00
ctrl_en	1	1	100.00
ctrl_loopback	1	1	100.00
ctrl_lsb_first	1	1	100.00
ctrl_mode[1-0]	1	1	100.00
ctrl_mstr	1	1	100.00
ctrl_width[1-0]	1	1	100.00
ctrl_word[0-7]	1	1	100.00
ctrl_word[8-31]	0	0	0.00
delay_cfg[7-0]	1	1	100.00
delay_word[0-7]	1	1	100.00
delay_word[8-31]	0	0	0.00
i[31-0]	0	0	0.00
int_en[4-0]	1	1	100.00
int_en_word[0-4]	1	1	100.00
int_en_word[5-31]	0	0	0.00
int_stat[4-0]	1	1	100.00
int_stat_word[0-4]	1	1	100.00

int_stat_word[5-31]	0	0	0.00
rx_count[0-3]	1	1	100.00
rx_empty_w	1	1	100.00
rx_full_w	1	1	100.00
rx_pop_this_cycle	1	1	100.00
rx_push_data[0-31]	1	1	100.00
rx_push_valid	1	1	100.00
rx_rp[3-0]	1	1	100.00
rx_wp[3-0]	1	1	100.00
ss_ctrl_word[0-7]	1	1	100.00
ss_ctrl_word[8-31]	0	0	0.00
ss_en[3-0]	1	1	100.00
ss_val[3-0]	1	1	100.00
status_word[0-6]	1	1	100.00
status_word[7-31]	0	0	0.00
transfer_done_pulse	1	1	100.00
tx_count[0-3]	1	1	100.00
tx_empty	1	1	100.00
tx_empty_w	1	1	100.00
tx_full_w	1	1	100.00
tx_pop	1	1	100.00
tx_push_accepted	1	1	100.00
tx_push_data[31-0]	1	1	100.00
tx_push_dropped	1	1	100.00
tx_push_valid	1	1	100.00
tx_rp[3-0]	1	1	100.00
tx_word[0-31]	1	1	100.00
tx_wp[3-0]	1	1	100.00

```
Total Node Count     =        558
Toggled Node Count   =        357
Untoggled Node Count =        201

Toggle Coverage      =      63.97% (714 of 1116 bins)

================================================================================
```

```
=
=== Instance: /spi_master_top/u_dut/u_core/u_core_sva
=== Design Unit: work.spi_core_sva
================================================================================
=

Assertion Coverage:
    Assertions                      16        16         0   100.00%
--------------------------------------------------------------------
Name                 File(Line)                   Failure      Pass
                                                  Count        Count
--------------------------------------------------------------------
/spi_master_top/u_dut/u_core/u_core_sva/assert__rx_push_single_cycle
                     ../tb/assertions/spi_sva.sv(475)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__xfer_done_pulse_single_cycle
                     ../tb/assertions/spi_sva.sv(469)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__bit_cnt_in_range
                     ../tb/assertions/spi_sva.sv(446)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__width_bits_consistent
                     ../tb/assertions/spi_sva.sv(440)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_miso_eff_loopback
                     ../tb/assertions/spi_sva.sv(311)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_miso_eff_normal
                     ../tb/assertions/spi_sva.sv(323)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_busy_state_cfg_r3
                     ../tb/assertions/spi_sva.sv(334)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_mosi_launch_edge
                     ../tb/assertions/spi_sva.sv(345)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_mosi_cpha1_first_launch
                     ../tb/assertions/spi_sva.sv(354)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_finish_generates_done_and_rxpush
                     ../tb/assertions/spi_sva.sv(366)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_finish_restores_idle_clock
                     ../tb/assertions/spi_sva.sv(375)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_keeps_idle_clock
                     ../tb/assertions/spi_sva.sv(386)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_resets_sclk_counter
                     ../tb/assertions/spi_sva.sv(394)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_decrements_gap_counter
                     ../tb/assertions/spi_sva.sv(402)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_to_idle_transition
                     ../tb/assertions/spi_sva.sv(410)
                                                        0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_increments_sclk_counter
                     ../tb/assertions/spi_sva.sv(418)
                                                        0          1

Directive Coverage:
    Directives                      11        11         0   100.00%
```

```
DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------
------------
Name                                    Design Design   Lang File(Line)
Hits Status
                                        Unit   UnitType

--------------------------------------------------------------------------------
------------
/spi_master_top/u_dut/u_core/u_core_sva/c_miso_eff_loopback
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(313)

20872 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_miso_eff_normal
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(325)

390 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_mosi_launch_edge
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(347)

416 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_mosi_cpha1_first_launch
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(356)

22 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_finish_generates_done_and_rxpush
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(368)

65 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_finish_restores_idle_clock
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(377)

65 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_keeps_idle_clock
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(388)

260 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_resets_sclk_counter
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(396)

130 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_decrements_gap_counter
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(404)

130 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_to_idle_transition
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(412)

3 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_increments_sclk_counter
                                        spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(420)

130 Covered
```

```
================================================================================
=
=== Instance: /spi_master_top/u_dut/u_core
=== Design Unit: work.spi_core
================================================================================
=
Branch Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        57        55         2    96.49%

================================Branch Details================================

Branch Coverage for instance /spi_master_top/u_dut/u_core

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ../rtl/spi_core.sv
------------------------------------IF
Branch------------------------------------
    80                                        21     Count coming in to IF
    80             1                           7         wire [5:0] width_bits =
(xfer_width == 2'b00) ? 6'd8  :
    81             1                          14
(xfer_width == 2'b01) ? 6'd16 : 6'd32;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    81                                        14     Count coming in to IF
    81             2                           7
(xfer_width == 2'b01) ? 6'd16 : 6'd32;
    81             3                           7
(xfer_width == 2'b01) ? 6'd16 : 6'd32;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    89                                       510     Count coming in to IF
    89             1                         435         wire miso_eff =
cfg_loopback ? MOSI : MISO;
    89             2                          75         wire miso_eff =
cfg_loopback ? MOSI : MISO;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    96                                       735     Count coming in to IF
    96             1                         224             if (lsb_first)
    98             1                         511             else
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    104                                       65     Count coming in to IF
    104            1                           8             align_rx = sh &
((total_bits == 6'd32) ? 32'hFFFF_FFFF :
    105            1                          57
((32'h1 << total_bits) - 32'h1));
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
```

```
    112                                     4905     Count coming in to IF
    112             1                         13             if (!PRESETn) begin
    130             1                       4892             end else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    135                                     4892     Count coming in to IF
    135             1                        106                 if (!cfg_en)
begin
    142             1                       4786                 end else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------CASE
Branch------------------------------------
    143                                     4786     Count coming in to CASE
    145             1                        313                         S_IDLE:
begin
    178             1                       4079
S_SHIFT: begin
    224             1                        134
S_FINISH: begin
    246             1                        260                         S_GAP:
begin
    259             1                    ***0***
default: state <= S_IDLE;
Branch totals: 4 hits of 5 branches = 80.00%

------------------------------------IF
Branch------------------------------------
    149                                      313     Count coming in to IF
    149             1                         68                             if
(!tx_empty && cfg_mstr && (ss_n_drive != 4'hF)) begin
                                             245     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    159                                       68     Count coming in to IF
    159             1                         52
bit_cnt    <= (cfg_width == 2'b00) ? 6'd8  :
    160             1                         16
(cfg_width == 2'b01) ? 6'd16 : 6'd32;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    160                                       16     Count coming in to IF
    160             2                          8
(cfg_width == 2'b01) ? 6'd16 : 6'd32;
    160             3                          8
(cfg_width == 2'b01) ? 6'd16 : 6'd32;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    165                                       68     Count coming in to IF
    165             1                         57
if (cfg_mode[0] == 1'b0) begin
                                              11     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
```

```
    166                                       57     Count coming in to IF
    166             1                          6
MOSI <= cfg_lsb_first ? tx_word[0]
    167             1                         51
: tx_word[
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    168                                       51     Count coming in to IF
    168             1                         46
((cfg_width==2'b00)?7:
    169             1                          5
(cfg_width==2'b01)?15:31)];
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    169                                        5     Count coming in to IF
    169             2                          3
(cfg_width==2'b01)?15:31)];
    169             3                          2
(cfg_width==2'b01)?15:31)];
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    179                                     4079     Count coming in to IF
    179             1                       1504                              if
(sclk_cnt == half_period - 1) begin
    218             1                       2575                              end
else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    189                                     1504     Count coming in to IF
    189             1                       1088
is_sample_edge = (cpha == 1'b0) ? leading : ~leading;
    189             2                        416
is_sample_edge = (cpha == 1'b0) ? leading : ~leading;
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    192                                     1504     Count coming in to IF
    192             1                        780
if (is_sample_edge) begin
                                             724     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    193                                      780     Count coming in to IF
    193             1                        224
if (xfer_lsb_first) begin
    195             1                        556
end else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    199                                      780     Count coming in to IF
    199             1                         65
```

```
if (bit_cnt == 6'd1) begin
                                             715     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    205                                     1504     Count coming in to IF
    205             1                        724
if (is_launch_edge) begin
                                             780     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    206                                      724     Count coming in to IF
    206             1                        724
if (bit_cnt > 6'd0) begin
                                         ***0***     All False Count
Branch totals: 1 hit of 2 branches = 50.00%

------------------------------------IF
Branch------------------------------------
    213                                     1504     Count coming in to IF
    213             1                         11
if (cpha == 1'b1 && leading && bit_cnt == width_bits) begin
                                            1493     All False Count
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    225                                      134     Count coming in to IF
    225             1                         65                             if
(sclk_cnt == half_period - 1) begin
    240             1                         69                             end
else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    234                                       65     Count coming in to IF
    234             1                          3
if (!tx_empty && cfg_delay != 8'h0) begin
    237             1                         62
end else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    248                                      260     Count coming in to IF
    248             1                        130                             if
(sclk_cnt == half_period - 1) begin
    254             1                        130                             end
else begin
Branch totals: 2 hits of 2 branches = 100.00%

------------------------------------IF
Branch------------------------------------
    250                                      130     Count coming in to IF
    250             1                          3
if (gap_cnt == 9'h1) begin
                                             127     All False Count
Branch totals: 2 hits of 2 branches = 100.00%
```

```
Condition Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Conditions                      19        17         2    89.47%

================================Condition
Details================================

Condition Coverage for instance /spi_master_top/u_dut/u_core --

  File ../rtl/spi_core.sv
----------------Focused Condition View-------------------
Line       80 Item    1  (xfer_width == 0)
Condition totals: 1 of 1 input term covered = 100.00%

         Input Term   Covered  Reason for no coverage   Hint
        -----------  --------  -----------------------  --------------
  (xfer_width == 0)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (xfer_width == 0)_0   -
  Row   2:          1  (xfer_width == 0)_1   -

----------------Focused Condition View-------------------
Line       81 Item    1  (xfer_width == 1)
Condition totals: 1 of 1 input term covered = 100.00%

         Input Term   Covered  Reason for no coverage   Hint
        -----------  --------  -----------------------  --------------
  (xfer_width == 1)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (xfer_width == 1)_0   -
  Row   2:          1  (xfer_width == 1)_1   -

----------------Focused Condition View-------------------
Line       104 Item    1  (total_bits == 32)
Condition totals: 1 of 1 input term covered = 100.00%

          Input Term   Covered  Reason for no coverage   Hint
         -----------  --------  -----------------------  --------------
  (total_bits == 32)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (total_bits == 32)_0  -
  Row   2:          1  (total_bits == 32)_1  -

----------------Focused Condition View-------------------
Line       149 Item    1  ((~tx_empty && cfg_mstr) && (ss_n_drive != 15))
Condition totals: 2 of 3 input terms covered = 66.66%

          Input Term   Covered  Reason for no coverage   Hint
         -----------  --------  -----------------------  --------------
            tx_empty         Y
            cfg_mstr         N  '_0' not hit             Hit '_0'
  (ss_n_drive != 15)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  tx_empty_0            ((ss_n_drive != 15) && cfg_mstr)
  Row   2:          1  tx_empty_1            -
```

```
  Row   3:    ***0***  cfg_mstr_0            ~tx_empty
  Row   4:          1  cfg_mstr_1            ((ss_n_drive != 15) && ~tx_empty)
  Row   5:          1  (ss_n_drive != 15)_0  (~tx_empty && cfg_mstr)
  Row   6:          1  (ss_n_drive != 15)_1  (~tx_empty && cfg_mstr)

----------------Focused Condition View-------------------
Line       159 Item    1  (cfg_width == 0)
Condition totals: 1 of 1 input term covered = 100.00%

        Input Term   Covered  Reason for no coverage   Hint
       -----------  --------  ----------------------  --------------
  (cfg_width == 0)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (cfg_width == 0)_0    -
  Row   2:          1  (cfg_width == 0)_1    -

----------------Focused Condition View-------------------
Line       160 Item    1  (cfg_width == 1)
Condition totals: 1 of 1 input term covered = 100.00%

        Input Term   Covered  Reason for no coverage   Hint
       -----------  --------  ----------------------  --------------
  (cfg_width == 1)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (cfg_width == 1)_0    -
  Row   2:          1  (cfg_width == 1)_1    -

----------------Focused Condition View-------------------
Line       179 Item    1  (sclk_cnt == (half_period - 1))
Condition totals: 1 of 1 input term covered = 100.00%

                       Input Term   Covered  Reason for no coverage   Hint
                      -----------  --------  ----------------------
--------------
  (sclk_cnt == (half_period - 1))         Y

     Rows:       Hits  FEC Target                          Non-masking
condition(s)
 ---------  ---------  --------------------
-------------------------
  Row   1:          1  (sclk_cnt == (half_period - 1))_0  -

  Row   2:          1  (sclk_cnt == (half_period - 1))_1  -


----------------Focused Condition View-------------------
Line       199 Item    1  (bit_cnt == 1)
Condition totals: 1 of 1 input term covered = 100.00%

      Input Term   Covered  Reason for no coverage   Hint
     -----------  --------  ----------------------  --------------
  (bit_cnt == 1)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (bit_cnt == 1)_0      -
  Row   2:          1  (bit_cnt == 1)_1      -

----------------Focused Condition View-------------------
Line       206 Item    1  (bit_cnt > 0)
```

```
Condition totals: 0 of 1 input term covered = 0.00%

     Input Term   Covered  Reason for no coverage   Hint
    -----------  --------  ----------------------  --------------
  (bit_cnt > 0)         N  '_0' not hit             Hit '_0'

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:    ***0***  (bit_cnt > 0)_0       -
  Row   2:          1  (bit_cnt > 0)_1       -

----------------Focused Condition View-------------------
Line       213 Item    1  ((cpha && leading) && (bit_cnt == width_bits))
Condition totals: 3 of 3 input terms covered = 100.00%

               Input Term   Covered  Reason for no coverage   Hint
              -----------  --------  ----------------------  --------------
                     cpha         Y
                  leading         Y
  (bit_cnt == width_bits)         Y

     Rows:       Hits  FEC Target                 Non-masking condition(s)
 ---------  ---------  --------------------       -------------------------
  Row   1:          1  cpha_0                     -
  Row   2:          1  cpha_1                     ((bit_cnt == width_bits) &&
leading)
  Row   3:          1  leading_0                  cpha
  Row   4:          1  leading_1                  ((bit_cnt == width_bits) &&
cpha)
  Row   5:          1  (bit_cnt == width_bits)_0  (cpha && leading)
  Row   6:          1  (bit_cnt == width_bits)_1  (cpha && leading)

----------------Focused Condition View-------------------
Line       225 Item    1  (sclk_cnt == (half_period - 1))
Condition totals: 1 of 1 input term covered = 100.00%

                       Input Term   Covered  Reason for no coverage   Hint
                      -----------  --------  ----------------------
--------------
  (sclk_cnt == (half_period - 1))         Y

     Rows:       Hits  FEC Target                         Non-masking
condition(s)
 ---------  ---------  --------------------
-------------------------
  Row   1:          1  (sclk_cnt == (half_period - 1))_0  -

  Row   2:          1  (sclk_cnt == (half_period - 1))_1  -


----------------Focused Condition View-------------------
Line       234 Item    1  (~tx_empty && (cfg_delay != 0))
Condition totals: 2 of 2 input terms covered = 100.00%

        Input Term   Covered  Reason for no coverage   Hint
       -----------  --------  ----------------------  --------------
          tx_empty         Y
  (cfg_delay != 0)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  tx_empty_0            (cfg_delay != 0)
  Row   2:          1  tx_empty_1            -
  Row   3:          1  (cfg_delay != 0)_0    ~tx_empty
```

```
  Row   4:          1  (cfg_delay != 0)_1    ~tx_empty

----------------Focused Condition View-------------------
Line       248 Item    1  (sclk_cnt == (half_period - 1))
Condition totals: 1 of 1 input term covered = 100.00%

                       Input Term   Covered  Reason for no coverage   Hint
                      -----------  --------  ----------------------
--------------
  (sclk_cnt == (half_period - 1))         Y

     Rows:       Hits  FEC Target                         Non-masking
condition(s)
 ---------  ---------  --------------------
-------------------------
  Row   1:          1  (sclk_cnt == (half_period - 1))_0  -

  Row   2:          1  (sclk_cnt == (half_period - 1))_1  -


----------------Focused Condition View-------------------
Line       250 Item    1  (gap_cnt == 1)
Condition totals: 1 of 1 input term covered = 100.00%

      Input Term   Covered  Reason for no coverage   Hint
     -----------  --------  ----------------------  --------------
  (gap_cnt == 1)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (gap_cnt == 1)_0      -
  Row   2:          1  (gap_cnt == 1)_1      -


Expression Coverage:
    Enabled Coverage              Bins   Covered    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Expressions                      9         9         0   100.00%

================================Expression
Details================================

Expression Coverage for instance /spi_master_top/u_dut/u_core --

  File ../rtl/spi_core.sv
----------------Focused Expression View-----------------
Line       87 Item    1  (state != S_IDLE)
Expression totals: 1 of 1 input term covered = 100.00%

         Input Term   Covered  Reason for no coverage   Hint
        -----------  --------  ----------------------  --------------
  (state != S_IDLE)         Y

     Rows:       Hits  FEC Target            Non-masking condition(s)
 ---------  ---------  --------------------  -------------------------
  Row   1:          1  (state != S_IDLE)_0   -
  Row   2:          1  (state != S_IDLE)_1   -

-----------Focused Expression View (Bimodal)------------
Line       89 Item    1  (cfg_loopback? MOSI: MISO)
Expression totals: 3 of 3 input terms covered = 100.00%

      Input Term   Covered  Reason for no coverage                  Hint
     -----------  --------  --------------------------------------
```

```
--------------
    cfg_loopback         Y
            MOSI         Y
            MISO         Y

    Rows:   Hits(->0)   Hits(->1)  FEC Target            Non-masking
condition(s)

---------  ----------  ----------  --------------------
-------------------------
 Row   1:           1           1  cfg_loopback_0        -

 Row   2:           1           0  cfg_loopback_1        -

 Row   3:           1           0  MOSI_0                cfg_loopback

 Row   4:           0           1  MOSI_1                cfg_loopback

 Row   5:           1           0  MISO_0                ~cfg_loopback

 Row   6:           0           1  MISO_1                ~cfg_loopback


-----------Focused Expression View (Bimodal)------------
Line       166 Item    1  (cfg_lsb_first? tx_word[0]: tx_word[cfg_width==0?
7:cfg_width==1?15:31])
Expression totals: 3 of 3 input terms covered = 100.00%

                                    Input Term   Covered  Reason for no coverage
Hint
                                   -----------  --------
--------------------------------------  --------------
                                 cfg_lsb_first         Y
                                    tx_word[0]         Y
    tx_word[cfg_width==0?7:cfg_width==1?15:31]         Y

    Rows:   Hits(->0)   Hits(->1)  FEC Target
Non-masking condition(s)

---------  ----------  ----------  --------------------
-------------------------
 Row   1:           0           1  cfg_lsb_first_0
-
 Row   2:           1           1  cfg_lsb_first_1
-
 Row   3:           1           0  tx_word[0]_0
cfg_lsb_first
 Row   4:           0           1  tx_word[0]_1
cfg_lsb_first
 Row   5:           1           0  tx_word[cfg_width==0?7:cfg_width==1?15:31]_0
~cfg_lsb_first
 Row   6:           0           1  tx_word[cfg_width==0?7:cfg_width==1?15:31]_1
~cfg_lsb_first

-----------Focused Expression View (Bimodal)------------
Line       189 Item    1  (~cpha? leading: ~leading)
Expression totals: 2 of 2 input terms covered = 100.00%

  Input Term   Covered  Reason for no coverage                 Hint
 -----------  --------  -------------------------------------  --------------
        cpha         Y
     leading         Y

    Rows:   Hits(->0)   Hits(->1)  FEC Target            Non-masking
```

```
condition(s)

---------  ----------  ----------  --------------------
-------------------------
 Row   1:          1          1  cpha_0                -

 Row   2:          1          0  cpha_1                -

 Row   3:          1          0  leading_0             ~cpha, cpha

 Row   4:          0          1  leading_1             ~cpha, cpha


FSM Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    FSM States                       4         4         0   100.00%
    FSM Transitions                  6         6         0   100.00%

================================FSM Details================================

FSM Coverage for instance /spi_master_top/u_dut/u_core --

FSM_ID: state
    Current State Object : state
    ----------------------
    State Value MapInfo :
    ---------------------
Line          State Name               Value
----          ----------               -----
 145              S_IDLE                   0
 178             S_SHIFT                   1
 224            S_FINISH                   2
 246               S_GAP                   3
    Covered States :
    ----------------
                   State           Hit_count
                   -----           ---------
                  S_IDLE                 184
                 S_SHIFT                  68
                S_FINISH                  65
                   S_GAP                   3
    Covered Transitions :
    ---------------------
Line            Trans_ID           Hit_count          Transition
----            --------           ---------          ----------
 172                   0                  68          S_IDLE -> S_SHIFT

 200                   1                  65          S_SHIFT -> S_FINISH

 136                   2                   3          S_SHIFT -> S_IDLE

 238                   3                  62          S_FINISH -> S_IDLE

 236                   4                   3          S_FINISH -> S_GAP

 251                   5                   3          S_GAP -> S_IDLE


    Summary                       Bins      Hits    Misses  Coverage
    -------                       ----      ----    ------  --------
        FSM States                   4         4         0   100.00%
```

```
        FSM Transitions              6         6         0   100.00%
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                      77        76         1    98.70%

================================Statement
Details================================

Statement Coverage for instance /spi_master_top/u_dut/u_core --

    Line         Item                      Count     Source
    ----         ----                      -----     ------
  File ../rtl/spi_core.sv
    19                                               module spi_core (
    20                                                   input  wire
PCLK,
    21                                                   input  wire
PRESETn,
    22
    23                                                   // Configuration (live
from regfile)
    24                                                   input  wire
cfg_en,
    25                                                   input  wire
cfg_mstr,
    26                                                   input  wire [1:0]
cfg_mode,
    27                                                   input  wire
cfg_lsb_first,
    28                                                   input  wire
cfg_loopback,
    29                                                   input  wire [1:0]
cfg_width,
    30                                                   input  wire [15:0]
cfg_clk_div,
    31                                                   input  wire [7:0]
cfg_delay,
    32
    33                                                   // SS observation: core
starts only when at least one SS lane is asserted
    34                                                   // low on the pins
(regfile owns the final drive).
    35                                                   input  wire [3:0]
ss_n_drive,
    36
    37                                                   // TX FIFO -> core
    38                                                   input  wire [31:0]
tx_word,
    39                                                   input  wire
tx_empty,
    40                                                   output reg
tx_pop,
    41
    42                                                   // core -> RX FIFO
    43                                                   output reg
rx_push_valid,
    44                                                   output reg  [31:0]
rx_push_data,
    45
    46                                                   // Status
    47                                                   output wire
busy,
    48                                                   output reg
```

```
transfer_done_pulse,

        // SPI pins
        output reg SCLK,
        output reg MOSI,
        input  wire MISO
    );

    // ----------------------------------------------------------------------
    // FSM
    // ----------------------------------------------------------------------
    typedef enum logic [1:0] {
        S_IDLE   = 2'd0,
        S_SHIFT  = 2'd1,
        S_FINISH = 2'd2,
        S_GAP    = 2'd3
    } xfer_state_e;

    xfer_state_e state;

    reg [1:0]  xfer_mode;
    reg        xfer_lsb_first;
    reg [1:0]  xfer_width;
    reg [15:0] xfer_div;

    reg [31:0] sh_tx;
    reg [31:0] sh_rx;
    reg [5:0]  bit_cnt;
    reg [16:0] sclk_cnt;
    reg [8:0]  gap_cnt;
    reg        sclk_phase;

1    22    wire [5:0] width_bits = (xfer_width == 2'b00) ? 6'd8  :
                                   (xfer_width == 2'b01) ? 6'd16 : 6'd32;

    wire cpol = xfer_mode[1];
    wire cpha = xfer_mode[0];
1    16    wire [16:0] half_period = {1'b0, xfer_div} + 17'd1;

1   206    assign busy = (state != S_IDLE);

1   513    wire miso_eff = cfg_loopback ? MOSI : MISO;

    // Helper functions
    function automatic logic get_tx_bit(input logic [31:0] v,
                                        input logic [5:0]  remaining,
                                        input logic [5:0]  total_bits,
```

```
 95                                              input logic        lsb_first);
 96                                                  if (lsb_first)
 97      1                   224                         get_tx_bit = v[total_bits - remaining];
 98                                                  else
 99      1                   511                         get_tx_bit = v[remaining - 1];
100                                              endfunction
101
102                                              function automatic logic [31:0] align_rx(input logic [31:0] sh,
103                                              input logic [5:0]  total_bits);
104      1                    65                     align_rx = sh & ((total_bits == 6'd32) ? 32'hFFFF_FFFF :
105                                              ((32'h1 << total_bits) - 32'h1));
106                                              endfunction
107
108                                              // ---------------------------------------------------------------------------
109                                              // FSM + datapath
110                                              // ---------------------------------------------------------------------------
111      1                  4905                 always @(posedge PCLK or negedge PRESETn) begin
112                                                  if (!PRESETn) begin
113      1                    13                         state <= S_IDLE;
114      1                    13                         SCLK <= 1'b0;
115      1                    13                         MOSI <= 1'b0;
116      1                    13                         sh_tx <= 32'h0;
117      1                    13                         sh_rx <= 32'h0;
118      1                    13                         bit_cnt <= 6'h0;
119      1                    13                         sclk_cnt <= 17'h0;
120      1                    13                         sclk_phase <= 1'b0;
121      1                    13                         gap_cnt <= 9'h0;
122      1                    13                         xfer_mode <= 2'b00;
123      1                    13                         xfer_lsb_first <= 1'b0;
124      1                    13                         xfer_width <= 2'b00;
125      1                    13                         xfer_div <= 16'h0;
126      1                    13                         tx_pop <= 1'b0;
127      1                    13                         rx_push_valid <= 1'b0;
128      1                    13                         rx_push_data <= 32'h0;
129      1                    13                         transfer_done_pulse <= 1'b0;
130                                                  end else begin
131      1                  4892                         tx_pop
```

```
<= 1'b0;
    132           1                      4892                rx_push_valid
<= 1'b0;
    133           1                      4892
transfer_done_pulse <= 1'b0;
    134
    135                                                      if (!cfg_en)
begin
    136           1                       106                    state
<= S_IDLE;
    137           1                       106                    SCLK
<= cfg_mode[1];   // hold at CPOL idle
    138           1                       106                    MOSI
<= 1'b0;
    139           1                       106                    sclk_cnt
<= 17'h0;
    140           1                       106                    sclk_phase
<= 1'b0;
    141           1                       106                    gap_cnt
<= 9'h0;
    142                                                      end else begin
    143                                                          case
(state)
    144                                                              //
---------------- IDLE ------------------------------------
    145                                                              S_IDLE:
begin
    146           1                       313
SCLK <= cfg_mode[1];
    147                                                                  //
Start condition: TX has data, master mode, and at
    148                                                                  //
least one SS lane on the pins is driven low.
    149                                                                  if
(!tx_empty && cfg_mstr && (ss_n_drive != 4'hF)) begin
    150
// Latch config for this transfer (R25)
    151           1                        68
xfer_mode      <= cfg_mode;
    152           1                        68
xfer_lsb_first <= cfg_lsb_first;
    153           1                        68
xfer_width     <= cfg_width;
    154           1                        68
xfer_div       <= cfg_clk_div;
    155
    156           1                        68
sh_tx  <= tx_word;
    157           1                        68
tx_pop <= 1'b1;
    158
    159           1                        68
bit_cnt    <= (cfg_width == 2'b00) ? 6'd8  :
    160
(cfg_width == 2'b01) ? 6'd16 : 6'd32;
    161           1                        68
sclk_cnt   <= 17'h0;
    162           1                        68
sclk_phase <= 1'b0;
    163
    164
// Present first bit immediately for CPHA=0
    165
if (cfg_mode[0] == 1'b0) begin
```

```
    166             1                         57
MOSI <= cfg_lsb_first ? tx_word[0]
    167
: tx_word[
    168
((cfg_width==2'b00)?7:
    169
(cfg_width==2'b01)?15:31)];
    170
end
    171             1                         68
sh_rx <= 32'h0;
    172             1                         68
state <= S_SHIFT;
    173             1                         68
SCLK  <= cfg_mode[1];
    174                                                                               end
    175                                                                           end
    176
    177                                                                           //
---------------- SHIFT ----------------------------------
    178
S_SHIFT: begin
    179                                                                               if
(sclk_cnt == half_period - 1) begin
    180             1                       1504
sclk_cnt   <= 17'h0;
    181             1                       1504
sclk_phase <= ~sclk_phase;
    182             1                       1504
SCLK       <= ~SCLK;
    183
    184
begin : edge_work
    185
logic leading;
    186
logic is_sample_edge;
    187
logic is_launch_edge;
    188             1                       1504
leading = ~sclk_phase;
    189             1                       1504
is_sample_edge = (cpha == 1'b0) ? leading : ~leading;
    190             1                       1504
is_launch_edge = ~is_sample_edge;
    191
    192
if (is_sample_edge) begin
    193
if (xfer_lsb_first) begin
    194             1                        224
sh_rx[width_bits - bit_cnt] <= miso_eff;
    195
end else begin
    196             1                        556
sh_rx[bit_cnt - 1] <= miso_eff;
    197
end
    198
    199
if (bit_cnt == 6'd1) begin
    200             1                         65
state <= S_FINISH;
```

```
    201
end
    202            1                        780
bit_cnt <= bit_cnt - 6'd1;
    203
end
    204
    205
if (is_launch_edge) begin
    206
if (bit_cnt > 6'd0) begin
    207            1                        724
MOSI <= get_tx_bit(sh_tx, bit_cnt,
    208
width_bits, xfer_lsb_first);
    209
end
    210
end
    211
    212
// CPHA=1 first-bit-launch on very first edge
    213
if (cpha == 1'b1 && leading && bit_cnt == width_bits) begin
    214            1                         11
MOSI <= get_tx_bit(sh_tx, bit_cnt,
    215
width_bits, xfer_lsb_first);
    216
end
    217
end
    218                                                                                  end
else begin
    219            1                       2575
sclk_cnt <= sclk_cnt + 17'h1;
    220                                                                                  end
    221                                                                              end
    222
    223                                                                              //
---------------- FINISH ----------------------------------
    224
S_FINISH: begin
    225                                                                                  if
(sclk_cnt == half_period - 1) begin
    226            1                         65
sclk_cnt   <= 17'h0;
    227            1                         65
SCLK       <= cpol;
    228            1                         65
sclk_phase <= 1'b0;
    229
    230            1                         65
rx_push_valid <= 1'b1;
    231            1                         65
rx_push_data  <= align_rx(sh_rx, width_bits);
    232            1                         65
transfer_done_pulse <= 1'b1;
    233
    234
if (!tx_empty && cfg_delay != 8'h0) begin
    235            1                          3
gap_cnt <= {1'b0, cfg_delay};
    236            1                          3
```

```
state   <= S_GAP;
    237
end else begin
    238            1                        62
state <= S_IDLE;
    239
end
    240                                                                               end
else begin
    241            1                        69
sclk_cnt <= sclk_cnt + 17'h1;
    242                                                                               end
    243                                                                           end
    244
    245                                                                           //
---------------- GAP -------------------------------------
    246                                                                           S_GAP:
begin
    247            1                       260
SCLK <= cpol;
    248                                                                               if
(sclk_cnt == half_period - 1) begin
    249            1                       130
sclk_cnt <= 17'h0;
    250
if (gap_cnt == 9'h1) begin
    251            1                         3
state <= S_IDLE;
    252
end
    253            1                       130
gap_cnt <= gap_cnt - 9'h1;
    254                                                                               end
else begin
    255            1                       130
sclk_cnt <= sclk_cnt + 17'h1;
    256                                                                               end
    257                                                                           end
    258
    259            1                   ***0***
default: state <= S_IDLE;

Toggle Coverage:
    Enabled Coverage             Bins      Hits    Misses  Coverage
    ----------------             ----      ----    ------  --------
    Toggles                       512       451        61    88.08%

================================Toggle Details================================

Toggle Coverage for instance /spi_master_top/u_dut/u_core --

                                              Node      1H->0L      0L->1H
"Coverage"

---------------------------------------------------------------
                                              MISO           0           1
50.00
                                              MOSI           1           1
100.00
                                              PCLK           1           1
100.00
                                           PRESETn           1           1
100.00
                                              SCLK           1           1
```

100.00	bit_cnt[0-5]	1	1
100.00	busy	1	1
100.00	cfg_clk_div[0-15]	1	1
100.00	cfg_delay[0-7]	1	1
100.00	cfg_en	1	1
100.00	cfg_loopback	1	1
100.00	cfg_lsb_first	1	1
100.00	cfg_mode[0-1]	1	1
100.00	cfg_mstr	1	1
100.00	cfg_width[0-1]	1	1
100.00	cpha	1	1
100.00	cpol	1	1
100.00	gap_cnt[0-7]	1	1
100.00	gap_cnt[8]	0	0
0.00	half_period[0-2]	1	1
100.00	half_period[3-7]	0	0
0.00	half_period[8]	1	1
100.00	half_period[9]	0	0
0.00	half_period[10]	1	1
100.00	half_period[11-16]	0	0
0.00	miso_eff	1	1
100.00	rx_push_data[0-31]	1	1
100.00	rx_push_valid	1	1
100.00	sclk_cnt[0-9]	1	1
100.00	sclk_cnt[10-16]	0	0
0.00	sclk_phase	1	1
100.00	sh_rx[0-31]	1	1
100.00	sh_tx[0-31]	1	1
100.00	ss_n_drive[0-3]	1	1
100.00	state	ENUM type	

Value	Count
50	100.00

S_IDLE

S_SHIFT

```
50      100.00
                                                                          S_FINISH
48      100.00
                                                                             S_GAP
1      100.00
                                        transfer_done_pulse           1           1
100.00
                                                   tx_empty           1           1
100.00
                                                     tx_pop           1           1
100.00
                                              tx_word[0-31]           1           1
100.00
                                            width_bits[0-2]           0           0
0.00
                                            width_bits[3-5]           1           1
100.00
                                              xfer_div[0-7]           1           1
100.00
                                              xfer_div[8-9]           0           0
0.00
                                               xfer_div[10]           1           1
100.00
                                            xfer_div[11-15]           0           0
0.00
                                             xfer_lsb_first           1           1
100.00
                                             xfer_mode[0-1]           1           1
100.00
                                            xfer_width[0-1]           1           1
100.00

Total Node Count     =        258
Toggled Node Count   =        227
Untoggled Node Count =         31

Toggle Coverage      =      88.08% (451 of 512 bins)

================================================================================
=
=== Instance: /spi_master_top/u_dut
=== Design Unit: work.spi_master
================================================================================
=
Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                        384       379         5    98.69%

================================Toggle Details================================

Toggle Coverage for instance /spi_master_top/u_dut --

                                                       Node      1H->0L      0L->1H
"Coverage"

---------------------------------------------------------------
                                                        IRQ           1           1
100.00
                                                       MISO           0           1
50.00
                                                       MOSI           1           1
100.00
                                                 PADDR[0-7]           1           1
```

```
100.00
                                          PCLK           1           1
100.00
                                       PENABLE           1           1
100.00
                                  PRDATA[0-31]           1           1
100.00
                                        PREADY           0           0
0.00
                                       PRESETn           1           1
100.00
                                          PSEL           1           1
100.00
                                       PSLVERR           0           0
0.00
                                  PWDATA[0-31]           1           1
100.00
                                        PWRITE           1           1
100.00
                                          SCLK           1           1
100.00
                                     SS_n[0-3]           1           1
100.00
                                          busy           1           1
100.00
                             cfg_clk_div[0-15]           1           1
100.00
                                cfg_delay[0-7]           1           1
100.00
                                        cfg_en           1           1
100.00
                                  cfg_loopback           1           1
100.00
                                 cfg_lsb_first           1           1
100.00
                                 cfg_mode[0-1]           1           1
100.00
                                      cfg_mstr           1           1
100.00
                                cfg_width[0-1]           1           1
100.00
                            rx_push_data[0-31]           1           1
100.00
                                 rx_push_valid           1           1
100.00
                                 ss_n_int[0-3]           1           1
100.00
                           transfer_done_pulse           1           1
100.00
                                      tx_empty           1           1
100.00
                                        tx_pop           1           1
100.00
                                 tx_word[0-31]           1           1
100.00

Total Node Count     =        192
Toggled Node Count   =        189
Untoggled Node Count =          3

Toggle Coverage      =      98.69% (379 of 384 bins)

================================================================================
=
```

```
=== Instance: /spi_master_top
=== Design Unit: work.spi_master_top
================================================================================
=
Statement Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Statements                       6         6         0   100.00%

================================Statement
Details================================

Statement Coverage for instance /spi_master_top --

    Line         Item                      Count     Source 
    ----         ----                      -----     ------ 
  File ../tb/tb_top.sv
    18                                               module spi_master_top;
    19                                                   import uvm_pkg::*;
    20                                                   import
spi_master_pkg::*;
    21                                                   `include
"uvm_macros.svh"
    22                                                   `timescale 1ns/1ps
    23                                                   // Clock Generation
    24                                                   bit clk ;
    25                                                   initial begin
    26              1                          1             forever begin
    27              1                      42552                 #1;
    28              1                      42551                 clk=!clk;
    29                                                       end
    30                                                   end
    31                                                  // instantite dut &
interfaces
    32                                                   spi_if spi_if(clk);
    33                                                   apb_if apb_if(clk);
    34                                                   
    35                                                   spi_master
u_dut(.PRESETn(apb_if.PRESETn),

36                                                              .PCLK(spi_if.PCL
K),

37                                                              .SCLK(spi_if.SCL
K),

38                                                              .MOSI(spi_if.MOS
I),

39                                                              .MISO(spi_if.MIS
O),

40                                                              .SS_n(spi_if.SS_
n),

41                                                              .IRQ(spi_if.IRQ)
,

42                                                              .PSEL(apb_if.PSE
L),

43                                                              .PENABLE(apb_if.
PENABLE),
```

```
        .PWRITE(apb_if.PWRITE),
        .PADDR(apb_if.PADDR),
        .PWDATA(apb_if.PWDATA),
        .PRDATA(apb_if.PRDATA),
        .PREADY(apb_if.PREADY),
        .PSLVERR(apb_if.PSLVERR)
    );


// =============================================================================
// BIND DECLARATIONS
// =============================================================================
bind apb_regfile apb_regfile_sva u_apb_sva (
    .PCLK (PCLK),
    .PRESETn (PRESETn),
    .PSEL (PSEL),
    .PENABLE (PENABLE),
    .PWRITE (PWRITE),
    .PADDR (PADDR),
    .PWDATA (PWDATA),
    .PRDATA (PRDATA),
    .PREADY (PREADY),
    .PSLVERR (PSLVERR),
    .ctrl_en (ctrl_en),
    .int_stat (int_stat),
    .int_en (int_en),
    .IRQ (IRQ),
    .rx_full_w (rx_full_w),
    .rx_push_valid (rx_push_valid),
    .tx_push_dropped (tx_push_dropped),
    .tx_push_accepted (tx_push_accepted),
    .tx_full_w
```

```
(tx_full_w),
    .busy_in (busy_in)
);

bind spi_core spi_core_sva u_core_sva (
    .PCLK (PCLK),
    .PRESETn (PRESETn),
    .cfg_en (cfg_en),
    .cfg_mstr (cfg_mstr),
    .cfg_mode (cfg_mode),
    .cfg_lsb_first (cfg_lsb_first),
    .cfg_loopback (cfg_loopback),
    .cfg_width (cfg_width),
    .cfg_clk_div (cfg_clk_div),
    .cfg_delay (cfg_delay),
    .SCLK (SCLK),
    .MOSI (MOSI),
    .MISO (MISO),
    .ss_n_drive (ss_n_drive),
    .tx_word (tx_word),
    .tx_empty (tx_empty),
    .tx_pop (tx_pop),
    .rx_push_valid (rx_push_valid),
    .rx_push_data (rx_push_data),
    .busy (busy),
    .transfer_done_pulse (transfer_done_pulse),
    .state (state),
    .bit_cnt (bit_cnt),
    .sclk_cnt (sclk_cnt),
    .gap_cnt (gap_cnt),
    .sclk_phase (sclk_phase),
    .sh_tx (sh_tx),
    .sh_rx (sh_rx),
    .xfer_mode
```

```
(xfer_mode),
    109                                                   .xfer_lsb_first
(xfer_lsb_first),
    110                                                   .xfer_width
(xfer_width),
    111                                                   .xfer_div
(xfer_div)
    112                                               );
    113
    114                                                   initial begin
    115
    116             1                          1
uvm_config_db#(virtual spi_if)::set(null,"uvm_test_top", "SPI_IF",   spi_if  );
    117             1                          1
uvm_config_db#(virtual apb_if)::set(null,"uvm_test_top", "APB_IF",   apb_if  );
    118
    119             1                          1
run_test("sanity_test");

Toggle Coverage:
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Toggles                          2         2         0   100.00%

================================Toggle Details================================

Toggle Coverage for instance /spi_master_top --

                                              Node      1H->0L      0L->1H
"Coverage"

---------------------------------------------------------------
                                               clk           1           1
100.00

Total Node Count     =          1
Toggled Node Count   =          1
Untoggled Node Count =          0

Toggle Coverage      =     100.00% (2 of 2 bins)

===============================================================================
=
=== Instance: /spi_master_pkg
=== Design Unit: work.spi_master_pkg
===============================================================================
=

Covergroup Coverage:
    Covergroups                      1        na        na   100.00%
        Coverpoints/Crosses         13        na        na        na
            Covergroup Bins         77        77         0   100.00%
-------------------------------------------------------------------------------
-------------------------
Covergroup                                           Metric       Goal
Bins    Status


-------------------------------------------------------------------------------
-------------------------
 TYPE /spi_master_pkg/spi_coverage/g1               100.00%        100
-    Covered
    covered/total bins:                                  77         77
-
```

```
    missing/total bins:                         0          77
-
    % Hit:                                100.00%         100
-
    Coverpoint cp_reg_address             100.00%         100
-    Covered
       covered/total bins:                      9           9
-
       missing/total bins:                      0           9
-
       % Hit:                             100.00%         100
-
       bin CTRL                               220           1
-    Covered
       bin STATUS                           20140           1
-    Covered
       bin TX_DATA                            202           1
-    Covered
       bin RX_DATA                            118           1
-    Covered
       bin CLK_DIV                            114           1
-    Covered
       bin SS_CTRL                            230           1
-    Covered
       bin INT_EN                              38           1
-    Covered
       bin INT_STAT                           170           1
-    Covered
       bin DELAY                               32           1
-    Covered
    Coverpoint cp_clk_div_value           100.00%         100
-    Covered
       covered/total bins:                      7           7
-
       missing/total bins:                      0           7
-
       % Hit:                             100.00%         100
-
       bin div_0                               10           1
-    Covered
       bin div_1                               60           1
-    Covered
       bin div_2                                4           1
-    Covered
       bin div_3                                6           1
-    Covered
       bin div_255                              6           1
-    Covered
       bin div_1024                             4           1
-    Covered
       bin div_65535                            2           1
-    Covered
    Coverpoint cp_lsb_first               100.00%         100
-    Covered
       covered/total bins:                      2           2
-
       missing/total bins:                      0           2
-
       % Hit:                             100.00%         100
-
       bin lsb_first                           24           1
-    Covered
       bin msb_first                          172           1
-    Covered
```

```
    Coverpoint cp_mode                                100.00%        100
-    Covered
       covered/total bins:                                  4          4
-
       missing/total bins:                                  0          4
-
       % Hit:                                         100.00%        100
-
       bin mode0                                          156          1
-    Covered
       bin mode1                                           16          1
-    Covered
       bin mode2                                           12          1
-    Covered
       bin mode3                                           12          1
-    Covered
    Coverpoint cp_width                               100.00%        100
-    Covered
       covered/total bins:                                  3          3
-
       missing/total bins:                                  0          3
-
       % Hit:                                         100.00%        100
-
       bin w8                                             160          1
-    Covered
       bin w16                                             18          1
-    Covered
       bin w32                                             18          1
-    Covered
    Coverpoint cp_delay_value                         100.00%        100
-    Covered
       covered/total bins:                                  3          3
-
       missing/total bins:                                  0          3
-
       % Hit:                                         100.00%        100
-
       bin delay_0                                         10          1
-    Covered
       bin delay_1                                          4          1
-    Covered
       bin delay_large                                      6          1
-    Covered
    Coverpoint cp_loop_back                           100.00%        100
-    Covered
       covered/total bins:                                  2          2
-
       missing/total bins:                                  0          2
-
       % Hit:                                         100.00%        100
-
       bin loopback_enabled                                82          1
-    Covered
       bin loopback_disabled                              114          1
-    Covered
    Coverpoint cp_pwrite                              100.00%        100
-    Covered
       covered/total bins:                                  2          2
-
       missing/total bins:                                  0          2
-
       % Hit:                                         100.00%        100
-
```

```
        bin write                                      856          1
-    Covered
        bin read                                     20420          1
-    Covered
    Coverpoint cp_prdata                           100.00%        100
-    Covered
        covered/total bins:                              2          2
-
        missing/total bins:                              0          2
-
        % Hit:                                     100.00%        100
-
        bin zero_val                                 10248          1
-    Covered
        bin status_val                                   1          1
-    Covered
    Cross cp_spi_mode_x_all_widths                 100.00%        100
-    Covered
        covered/total bins:                             24         24
-
        missing/total bins:                              0         24
-
        % Hit:                                     100.00%        100
-
        Auto, Default and User Defined Bins:
            bin mode1_with_w8_msb                      142          1
-    Covered
            bin mode1_with_w16_msb                       4          1
-    Covered
            bin mode1_with_w32_msb                       4          1
-    Covered
            bin mode2_with_w8_msb                        6          1
-    Covered
            bin mode2_with_w16_msb                       2          1
-    Covered
            bin mode2_with_w32_msb                       2          1
-    Covered
            bin mode3_with_w8_msb                        2          1
-    Covered
            bin mode3_with_w16_msb                       2          1
-    Covered
            bin mode3_with_w32_msb                       2          1
-    Covered
            bin mode4_with_w8_msb                        2          1
-    Covered
            bin mode4_with_w16_msb                       2          1
-    Covered
            bin mode4_with_w32_msb                       2          1
-    Covered
            bin mode1_with_w8_lsb                        2          1
-    Covered
            bin mode1_with_w16_lsb                       2          1
-    Covered
            bin mode1_with_w32_lsb                       2          1
-    Covered
            bin mode2_with_w8_lsb                        2          1
-    Covered
            bin mode2_with_w16_lsb                       2          1
-    Covered
            bin mode2_with_w32_lsb                       2          1
-    Covered
            bin mode3_with_w8_lsb                        2          1
-    Covered
            bin mode3_with_w16_lsb                       2          1
```

```
-    Covered
            bin mode3_with_w32_lsb                           2           1
-    Covered
            bin mode4_with_w8_lsb                            2           1
-    Covered
            bin mode4_with_w16_lsb                           2           1
-    Covered
            bin mode4_with_w32_lsb                           2           1
-    Covered
    Cross cp_clock_div_corners                         100.00%         100
-    Covered
        covered/total bins:                                  8           8
-
        missing/total bins:                                  0           8
-
        % Hit:                                         100.00%         100
-
        Auto, Default and User Defined Bins:
            bin div_0                                       10           1
-    Covered
            bin div_1                                       60           1
-    Covered
            bin div_2                                        4           1
-    Covered
            bin div_3                                        6           1
-    Covered
            bin div_255                                      6           1
-    Covered
            bin div_1024                                     4           1
-    Covered
            bin div_65535                                    2           1
-    Covered
            bin div_random                                  92           1
-    Covered
    Cross cp_each_reg_read_write                       100.00%         100
-    Covered
        covered/total bins:                                  2           2
-
        missing/total bins:                                  0           2
-
        % Hit:                                         100.00%         100
-
        Auto, Default and User Defined Bins:
            bin reg_read                                 20410           1
-    Covered
            bin reg_write                                  854           1
-    Covered
    Cross cp_reset_observed                            100.00%         100
-    Covered
        covered/total bins:                                  9           9
-
        missing/total bins:                                  0           9
-
        % Hit:                                         100.00%         100
-
        Auto, Default and User Defined Bins:
            bin ctrl_reset                                  20           1
-    Covered
            bin status_reset                                 1           1
-    Covered
            bin tx_data_reset                                2           1
-    Covered
            bin rx_data_reset                               63           1
-    Covered
```

```
            bin clk_div_reset                               11          1
-    Covered
            bin ss_ctrl_reset                               14          1
-    Covered
            bin int_en_reset                                 7          1
-    Covered
            bin int_stat_reset                              37          1
-    Covered
            bin delay_reset                                  6          1
-    Covered

COVERGROUP COVERAGE:
--------------------------------------------------------------------------------
--------------------------
Covergroup                                              Metric       Goal
Bins    Status


--------------------------------------------------------------------------------
--------------------------
 TYPE /spi_master_pkg/spi_coverage/g1                  100.00%        100
-    Covered
    covered/total bins:                                     77         77
-
    missing/total bins:                                      0         77
-
    % Hit:                                             100.00%        100
-
    Coverpoint cp_reg_address                          100.00%        100
-    Covered
        covered/total bins:                                  9          9
-
        missing/total bins:                                  0          9
-
        % Hit:                                         100.00%        100
-
        bin CTRL                                           220          1
-    Covered
        bin STATUS                                       20140          1
-    Covered
        bin TX_DATA                                        202          1
-    Covered
        bin RX_DATA                                        118          1
-    Covered
        bin CLK_DIV                                        114          1
-    Covered
        bin SS_CTRL                                        230          1
-    Covered
        bin INT_EN                                          38          1
-    Covered
        bin INT_STAT                                       170          1
-    Covered
        bin DELAY                                           32          1
-    Covered
    Coverpoint cp_clk_div_value                        100.00%        100
-    Covered
        covered/total bins:                                  7          7
-
        missing/total bins:                                  0          7
-
        % Hit:                                         100.00%        100
-
        bin div_0                                           10          1
-    Covered
```

```
        bin div_1                                        60          1
-    Covered
        bin div_2                                         4          1
-    Covered
        bin div_3                                         6          1
-    Covered
        bin div_255                                       6          1
-    Covered
        bin div_1024                                      4          1
-    Covered
        bin div_65535                                     2          1
-    Covered
    Coverpoint cp_lsb_first                         100.00%        100
-    Covered
        covered/total bins:                               2          2
-
        missing/total bins:                               0          2
-
        % Hit:                                      100.00%        100
-
        bin lsb_first                                    24          1
-    Covered
        bin msb_first                                   172          1
-    Covered
    Coverpoint cp_mode                              100.00%        100
-    Covered
        covered/total bins:                               4          4
-
        missing/total bins:                               0          4
-
        % Hit:                                      100.00%        100
-
        bin mode0                                       156          1
-    Covered
        bin mode1                                        16          1
-    Covered
        bin mode2                                        12          1
-    Covered
        bin mode3                                        12          1
-    Covered
    Coverpoint cp_width                             100.00%        100
-    Covered
        covered/total bins:                               3          3
-
        missing/total bins:                               0          3
-
        % Hit:                                      100.00%        100
-
        bin w8                                          160          1
-    Covered
        bin w16                                          18          1
-    Covered
        bin w32                                          18          1
-    Covered
    Coverpoint cp_delay_value                       100.00%        100
-    Covered
        covered/total bins:                               3          3
-
        missing/total bins:                               0          3
-
        % Hit:                                      100.00%        100
-
        bin delay_0                                      10          1
-    Covered
```

```
        bin delay_1                                     4          1
-    Covered
        bin delay_large                                 6          1
-    Covered
    Coverpoint cp_loop_back                       100.00%        100
-    Covered
        covered/total bins:                             2          2
-
        missing/total bins:                             0          2
-
        % Hit:                                    100.00%        100
-
        bin loopback_enabled                           82          1
-    Covered
        bin loopback_disabled                         114          1
-    Covered
    Coverpoint cp_pwrite                          100.00%        100
-    Covered
        covered/total bins:                             2          2
-
        missing/total bins:                             0          2
-
        % Hit:                                    100.00%        100
-
        bin write                                     856          1
-    Covered
        bin read                                    20420          1
-    Covered
    Coverpoint cp_prdata                          100.00%        100
-    Covered
        covered/total bins:                             2          2
-
        missing/total bins:                             0          2
-
        % Hit:                                    100.00%        100
-
        bin zero_val                                10248          1
-    Covered
        bin status_val                                  1          1
-    Covered
    Cross cp_spi_mode_x_all_widths                100.00%        100
-    Covered
        covered/total bins:                            24         24
-
        missing/total bins:                             0         24
-
        % Hit:                                    100.00%        100
-
        Auto, Default and User Defined Bins:
            bin mode1_with_w8_msb                     142          1
-    Covered
            bin mode1_with_w16_msb                      4          1
-    Covered
            bin mode1_with_w32_msb                      4          1
-    Covered
            bin mode2_with_w8_msb                       6          1
-    Covered
            bin mode2_with_w16_msb                      2          1
-    Covered
            bin mode2_with_w32_msb                      2          1
-    Covered
            bin mode3_with_w8_msb                       2          1
-    Covered
            bin mode3_with_w16_msb                      2          1
```

```
-    Covered
            bin mode3_with_w32_msb                          2          1
-    Covered
            bin mode4_with_w8_msb                           2          1
-    Covered
            bin mode4_with_w16_msb                          2          1
-    Covered
            bin mode4_with_w32_msb                          2          1
-    Covered
            bin mode1_with_w8_lsb                           2          1
-    Covered
            bin mode1_with_w16_lsb                          2          1
-    Covered
            bin mode1_with_w32_lsb                          2          1
-    Covered
            bin mode2_with_w8_lsb                           2          1
-    Covered
            bin mode2_with_w16_lsb                          2          1
-    Covered
            bin mode2_with_w32_lsb                          2          1
-    Covered
            bin mode3_with_w8_lsb                           2          1
-    Covered
            bin mode3_with_w16_lsb                          2          1
-    Covered
            bin mode3_with_w32_lsb                          2          1
-    Covered
            bin mode4_with_w8_lsb                           2          1
-    Covered
            bin mode4_with_w16_lsb                          2          1
-    Covered
            bin mode4_with_w32_lsb                          2          1
-    Covered
    Cross cp_clock_div_corners                        100.00%        100
-    Covered
        covered/total bins:                                 8          8
-
        missing/total bins:                                 0          8
-
        % Hit:                                        100.00%        100
-
        Auto, Default and User Defined Bins:
            bin div_0                                      10          1
-    Covered
            bin div_1                                      60          1
-    Covered
            bin div_2                                       4          1
-    Covered
            bin div_3                                       6          1
-    Covered
            bin div_255                                     6          1
-    Covered
            bin div_1024                                    4          1
-    Covered
            bin div_65535                                   2          1
-    Covered
            bin div_random                                 92          1
-    Covered
    Cross cp_each_reg_read_write                      100.00%        100
-    Covered
        covered/total bins:                                 2          2
-
        missing/total bins:                                 0          2
-
```

```
        % Hit:                                        100.00%        100
-
        Auto, Default and User Defined Bins:
            bin reg_read                                20410          1
-    Covered
            bin reg_write                                 854          1
-    Covered
    Cross cp_reset_observed                           100.00%        100
-    Covered
        covered/total bins:                                 9          9
-
        missing/total bins:                                 0          9
-
        % Hit:                                        100.00%        100
-
        Auto, Default and User Defined Bins:
            bin ctrl_reset                                 20          1
-    Covered
            bin status_reset                                1          1
-    Covered
            bin tx_data_reset                               2          1
-    Covered
            bin rx_data_reset                              63          1
-    Covered
            bin clk_div_reset                              11          1
-    Covered
            bin ss_ctrl_reset                              14          1
-    Covered
            bin int_en_reset                                7          1
-    Covered
            bin int_stat_reset                             37          1
-    Covered
            bin delay_reset                                 6          1
-    Covered

TOTAL COVERGROUP COVERAGE: 100.00%  COVERGROUP TYPES: 1

DIRECTIVE COVERAGE:
--------------------------------------------------------------------------------
------------
Name                                     Design Design   Lang File(Line)
Hits Status
                                         Unit   UnitType

--------------------------------------------------------------------------------
------------
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__irq_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(212)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__int_stat_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(201)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__ctrl_en_reset_value
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(190)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_drop_implies_full
                                         apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(179)
```

```
6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_push_not_when_full
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(168)

21262 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__rx_ovf_sets_int_stat
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(157)

3 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__tx_ovf_sets_int_stat
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(146)

6 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__irq_equation_correct
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(135)

21262 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_pslverr_always_0
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(123)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_pready_always_1
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(112)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_ctrl_stable_setup_to_access

                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(101)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_penable_requires_psel
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(80)

10631 Covered
/spi_master_top/u_dut/u_regfile/u_apb_sva/cover__apb_psel_min_2_cycles
                                          apb_regfile_sva Verilog  SVA
../tb/assertions/spi_sva.sv(69)

6 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_miso_eff_loopback
                                          spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(313)

20872 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_miso_eff_normal
                                          spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(325)

390 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_mosi_launch_edge
                                          spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(347)

416 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_mosi_cpha1_first_launch
```

```
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(356)

22 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_finish_generates_done_and_rxpush
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(368)

65 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_finish_restores_idle_clock
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(377)

65 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_keeps_idle_clock
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(388)

260 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_resets_sclk_counter
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(396)

130 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_decrements_gap_counter
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(404)

130 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_to_idle_transition
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(412)

3 Covered
/spi_master_top/u_dut/u_core/u_core_sva/c_gap_increments_sclk_counter
                                         spi_core_sva Verilog  SVA
../tb/assertions/spi_sva.sv(420)

130 Covered

TOTAL DIRECTIVE COVERAGE: 100.00%  COVERS: 24

ASSERTION RESULTS:
--------------------------------------------------------------------
Name                 File(Line)                    Failure      Pass
                                                   Count        Count
--------------------------------------------------------------------
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__irq_reset_value
                     ../tb/assertions/spi_sva.sv(210)
                                                         0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__int_stat_reset_value
                     ../tb/assertions/spi_sva.sv(199)
                                                         0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__ctrl_en_reset_value
                     ../tb/assertions/spi_sva.sv(188)
                                                         0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_drop_implies_full
                     ../tb/assertions/spi_sva.sv(177)
                                                         0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_push_not_when_full
                     ../tb/assertions/spi_sva.sv(166)
                                                         0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__rx_ovf_sets_int_stat
                     ../tb/assertions/spi_sva.sv(155)
```

```
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__tx_ovf_sets_int_stat
                     ../tb/assertions/spi_sva.sv(144)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__irq_equation_correct
                     ../tb/assertions/spi_sva.sv(132)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_pslverr_always_0
                     ../tb/assertions/spi_sva.sv(121)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_pready_always_1
                     ../tb/assertions/spi_sva.sv(110)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/
assert__apb_ctrl_stable_setup_to_access
                     ../tb/assertions/spi_sva.sv(99)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_penable_requires_psel
                     ../tb/assertions/spi_sva.sv(78)
                                                                  0          1
/spi_master_top/u_dut/u_regfile/u_apb_sva/assert__apb_psel_min_2_cycles
                     ../tb/assertions/spi_sva.sv(67)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__rx_push_single_cycle
                     ../tb/assertions/spi_sva.sv(475)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__xfer_done_pulse_single_cycle
                     ../tb/assertions/spi_sva.sv(469)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__bit_cnt_in_range
                     ../tb/assertions/spi_sva.sv(446)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/assert__width_bits_consistent
                     ../tb/assertions/spi_sva.sv(440)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_miso_eff_loopback
                     ../tb/assertions/spi_sva.sv(311)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_miso_eff_normal
                     ../tb/assertions/spi_sva.sv(323)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_busy_state_cfg_r3
                     ../tb/assertions/spi_sva.sv(334)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_mosi_launch_edge
                     ../tb/assertions/spi_sva.sv(345)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_mosi_cpha1_first_launch
                     ../tb/assertions/spi_sva.sv(354)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_finish_generates_done_and_rxpush
                     ../tb/assertions/spi_sva.sv(366)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_finish_restores_idle_clock
                     ../tb/assertions/spi_sva.sv(375)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_keeps_idle_clock
                     ../tb/assertions/spi_sva.sv(386)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_resets_sclk_counter
                     ../tb/assertions/spi_sva.sv(394)
                                                                  0          1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_decrements_gap_counter
                     ../tb/assertions/spi_sva.sv(402)
```

```
                                                                 0           1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_to_idle_transition
                         ../tb/assertions/spi_sva.sv(410)
                                                                 0           1
/spi_master_top/u_dut/u_core/u_core_sva/a_gap_increments_sclk_counter
                         ../tb/assertions/spi_sva.sv(418)
                                                                 0           1

Total Coverage By Instance (filtered view): 95.76%
```